AVAILABILITY MATH

# Four nines is a budget, not a grade.

# 52.56

## minutes of downtime a year at 99.99%

What the common availability targets actually allow.

Swipe for the full table →

THE ALLOWANCE

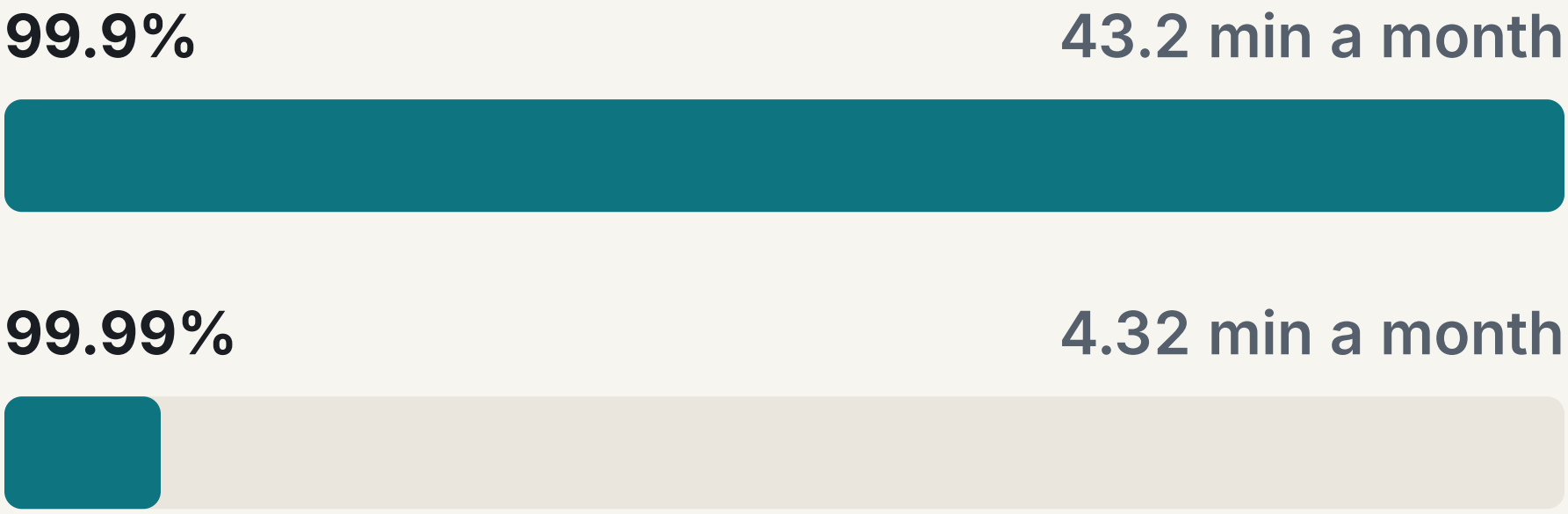
# The downtime budget behind each target

| TARGET | PER YEAR | PER 30-DAY MONTH |
|---|---|---|
| **99%** | 87.6 hours | 7.2 hours |
| **99.9%** | 8.76 hours | 43.2 minutes |
| **99.95%** | 4.38 hours | 21.6 minutes |
| **99.99%** | 52.56 minutes | 4.32 minutes |
| **99.999%** | 5 min 15 s | 26 seconds |

Assumes a 365-day year and a 30-day month. Seconds are rounded.

EACH EXTRA NINE

# Each nine leaves ten times less time

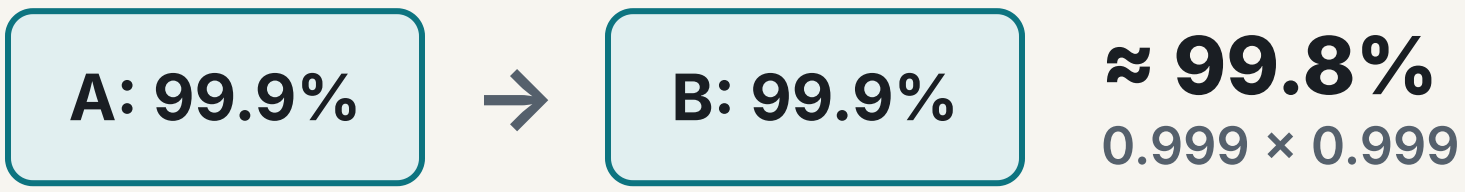



> At 99.99%, one 45-minute maintenance window uses about 86% of the whole year's allowance, before anything has broken.

DEPENDENCIES

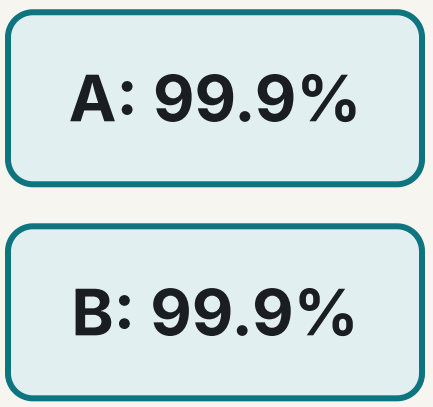
# Everything in the path shares the budget

## IN SERIES: BOTH MUST WORK




**≈ 99.8%**
0.999 × 0.999

## IN PARALLEL: EITHER CAN SERVE




**99.9999%**
on paper: 1 − 0.001 × 0.001

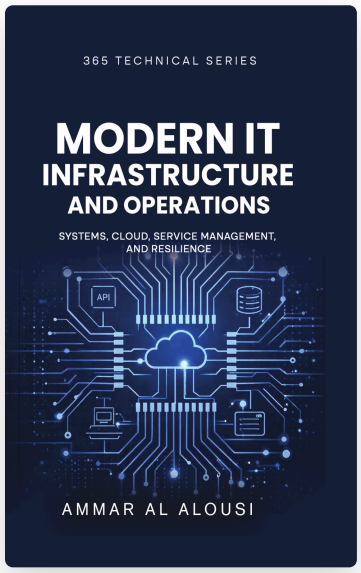
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**Only if failures are independent.** Shared power, network paths, configuration, or one deployment pipeline make them fail together.

BEFORE YOU COMMIT

# Translate the target into minutes first

Convert it to minutes per month. Then list who spends them: maintenance, deployments, failures in services you depend on, and the incident nobody has had yet.




Go deeper on resilience in

**Modern IT Infrastructure and Operations**

**Link in bio**

52.56 minutes: Google SRE book, “Embracing Risk” (sre.google). Other figures calculated from a 365-day year and 30-day month; parallel figure assumes independent failures.